OPPORTUNITY BRIEF · SALES AND PRESALES

LPWAN-based energy monitoring for maritime ports to optimize renewable energy integration and reduce operational costs.

Transportation and Logistics × Energy optimisation × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS071 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
43 is the world moving	**50** can we play, can we win	**MEDIUM** 0 named in the evidence	**€2.4bn** partial evidence · per year	**NOW** budgeted procurement within 1...	**L0** 2 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying LPWAN-based energy monitoring in maritime ports to optimize renewable energy integration and cut operational costs. Ports are increasingly adopting renewable energy sources, and real-time monitoring helps manage this transition efficiently. The time is right because ports are actively procuring environmental monitoring services, as seen in recent tenders.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	**€3.8bn** €831m – €22.2bn	**€2.4bn** €518m – €13.8bn	**€35.5m** €7.8m – €207m	partial
Observed: tendered contract value, annualised	€115m	€115m	€1.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M)	32.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only.

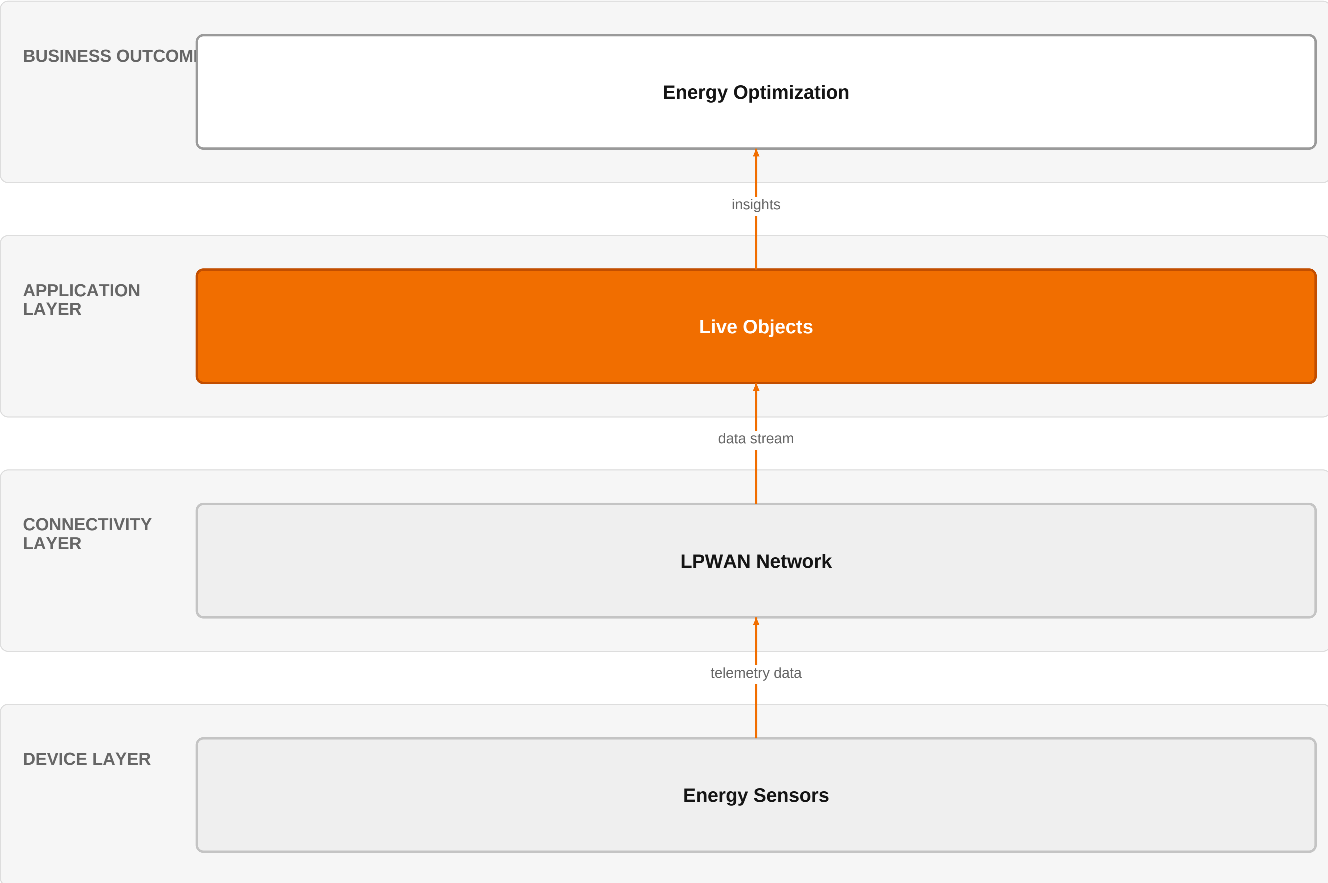
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (CA, US); the estimate covers the European footprint only.

The solution, in one picture

LPWAN Energy Monitoring for Ports



Orange asset · Partner · Customer-owned · Third party / to be sourced

This diagram shows how LPWAN sensors feed data into Orange's Live Objects platform to drive energy optimization decisions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Maritime ports are integrating renewable energy sources into their power distribution systems, driven by benefits such as environmental friendliness, cost-effective power generation, and energy sustainability. This shift increases the need for real-time monitoring to manage variable renewable generation and optimize energy use. Regulatory and procurement activity, such as the Norwegian environmental monitoring tender, indicates ports are actively seeking monitoring solutions.

`Sources: SIG-AA6F24D77461 ted.europa.eu 2026-05-27; SIG-F98C940013AB openalex.org 2025-08-22`

Who buys, and why now

The buyer is typically the port authority's head of operations or sustainability officer. The pain is operational: integrating renewables without real-time data leads to inefficiencies, higher costs, and potential grid instability. The trigger is often a new environmental regulation, a sustainability mandate, or a budgeted procurement cycle for environmental monitoring services.

Why it is hot now — every claim bound to a dated source

- Renewable energy sources are being integrated into maritime systems like ports, with benefits including environmental friendliness, cost-effective power generation, and energy sustainability. [SIG-F98C940013AB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Live Objects platform to provide end-to-end energy monitoring. The engagement would start with a discovery phase involving Orange Group researchers to assess the port's energy profile, then deploy LPWAN sensors (via partners) connected to Live Objects for data aggregation and analytics. Digital experts would integrate the data into the port's existing systems for actionable insights.

Why Orange can win

Orange can win by leveraging its Live Objects platform and IoT expertise to deliver a scalable, secure, and integrated monitoring solution. The combination of Orange Group researchers and defense/security experts adds credibility for handling critical infrastructure. However, the asset list is thin; Orange lacks a dedicated maritime energy solution, so differentiation must come from integration and trust.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.6; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom peers like Telia and Vodafone, who offer similar LPWAN connectivity, and industrial specialists like Actility and Honeywell, who bring deep OT expertise. Nokia and Boldyn Networks compete on network infrastructure, while Marlink offers satellite-based connectivity. Dedrone by Axon is a security specialist active in the sector. The customer will compare on price, reliability, and domain expertise.

`Sources: SIG-AA6F24D77461 ted.europa.eu 2026-05-27; SIG-F98C940013AB openalex.org 2025-08-22`

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); active in Transportation and Logistics; competes in Connectivity Solutions, OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Dedrone by Axon	Security specialist	active in Transportation and Logistics; competes in OX: Smart Industries	structural	—
Honeywell	Industrial / OT specialist	active in Transportation and Logistics; competes in OX: Smart Industries	structural	—
Boldyn Networks	Network equipment vendor	active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Marlink	Satellite operator	active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Nokia	Network equipment vendor	active in Transportation and Logistics; competes in Connectivity Solutions, OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

1. How are you currently monitoring energy consumption from renewable sources at your port?
2. What is the biggest operational challenge you face with integrating renewables into your power grid?
3. Have you issued a tender or budgeted for environmental monitoring services in the next 12 months?
4. Who in your organization is responsible for sustainability and energy management?
5. What existing IoT or SCADA systems do you have in place, and how do they integrate with your energy management?

Objections you will meet

They will say	You can answer
We already have a monitoring system in place.	That's a fair point. Many ports have legacy systems. Our solution is designed to complement and enhance existing infrastructure, providing a unified view across all energy sources.
LPWAN coverage in our port is unreliable.	Coverage is a valid concern. We can conduct a site survey to assess coverage and design a network that ensures reliability, possibly using a mix of LPWAN technologies.
We are concerned about data security and sovereignty.	Security is paramount. Orange has defense and security experts who can ensure the solution meets sovereign and certified standards, and our Live Objects platform is built with security in mind.

Risks and unknowns

The main risk is that ports may prefer a specialist with proven maritime energy monitoring deployments. Unknowns include the specific regulatory requirements per port and the willingness to invest in new IoT infrastructure. The business case depends on energy savings, which are not yet quantified for this use case.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a maritime port energy monitoring opportunity, anchoring the proposal on Orange's Live Objects platform and the expertise of Orange IoT and Digital experts, and referencing Point One Navigation as a proof point of Orange's ability to deliver precise, reliable IoT solutions.
Sales	In a conversation with the head of facilities or energy manager at a European port authority, open with: 'We've seen ports increasingly turn to renewable energy to cut costs and meet sustainability goals, and we're exploring how LPWAN-based monitoring can help integrate those sources more efficiently. Could we share what we're learning and see if it resonates with your current challenges?'
Strategist / Innovator	Commission a deep-dive brief on LPWAN-based energy monitoring for maritime ports, leveraging Orange Group researchers and IoT experts to map the technical and commercial landscape, and to identify where Orange's Live Objects platform can be positioned against incumbent solutions.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AA6F24D77461	2026-05-27	ted.europa.eu	1	Norway – Environmental monitoring other than for construction – Monitoring foreign marine speci...
SIG-F98C940013AB	2025-08-22	openalex.org	1	Smart, Connected, and Sustainable: The Transformation of Maritime Ports Through Electrification...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:39+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.